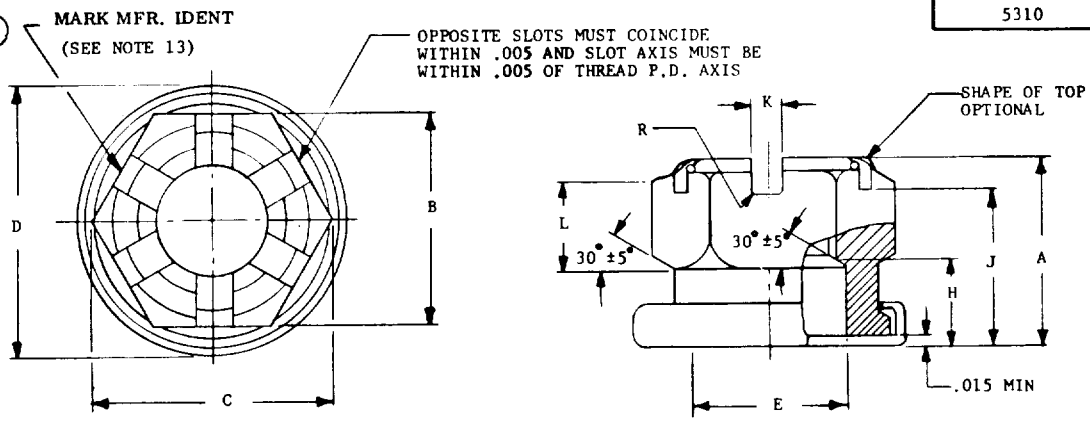
PROJECT NO. 5310-0876

FED. SUP CLASS 5310



SIZE DASH NO. △	(b) THREADS PER MIL-S-8879	A		B		C	D		E (c)	E (d)	H	
		MAX	MIN	MAX	MIN	MIN	MAX	MIN	MAX	MIN	MAX	MIN
-3	.1900.32UNJF.3B	.374	.334	.376	.367	.413	.490	.460	.240	.230	.161	.141
-4	.2500-28UNJF-3B	.405	.365	.439	.430	.488	.560	.530	.309	.299	.192	.172
-5	.3125-24UNJF-3B	.436	.396	.502	.492	.557	.640	.610	.395	.385	.223	.203
-6	.3750-24UNJF-3B	.482	.442	.564	.553	.628	.710	.680	.460	.450	.238	.218
-7	.4375-20UNJF-3B	.513	.473	.627	.616	.698	.785	.755	.529	.519	.269	.249
-8	.5000-20UNJF-3B	.576	.536	.752	.741	.840	.925	.895	.614	.604	.301	.281
-9	.5625-18UNJF-3B	.670	.630	.877	.865	.988	1.080	1.050	.686	.676	.332	.312
-10	.6250-18UNJF-3B	.822	.792	1.002	.990	1.121	1.224	1.194	.770	.760	.407	.387
-12	.7500-16UNJF-3B	.951	.911	1.064	1.052	1.191	1.283	1.253	.915	.905	.450	.430
-14	.8750-14UNJF-3B	1.058	1.018	1.252	1.239	1.403	1.495	1.465	1.065	1.055	.495	.475
-16	1.0000-12UNJF-3B	1.202	1.162	1.440	1.427	1.615	1.707	1.677	1.215	1.205	.544	.524

SIZE DASH NO. △	J		K		L	R	X WRENCH TORQUE CAPABILITY	Y INSTL. TORQUE	Z LOCKING TORQUE	ULTIMATE TENSILE STRENGTH (a)	WEIGHT
	MAX	MIN	MAX	MIN	MIN	MAX	MIN IN-LB	MAX IN-LB	MAX IN-LB	LBS MIN	LB/100 MAX
- 3	.279	.259	.108	.070	.110	.035	30	20	9	1,000	.91
- 4	.316	.296					65	45	15	1,800	1.20
- 5	.347	.327		.078	.125		115	90	25	2,900	1.50
- 6	.378	.358	.155	.125	.140	.053	200	125	40	4,100	1.90
- 7	.409	.389			.130		300	225	50	5,000	2.17
- 8	.472	.452			.150		375	300	75	5,600	3.60
- 9	.550	.530	.187	.157	.165	.062	500	400	100	7,800	6.20
-10	.633	.613			.170		750	600	150	10,000	6.90
-12	.765	.745			.220		1400	1100	200	15,500	7.60
-14	.871	.851			.288		2400	1900	300	22,800	13.70
-16	.984	.964			.333		3800	3000	400	35,500	17.10

REVIEWER SYMBOLS: NAVY - AS, USAF - 11, ARMY-AV, EL, DSA - IS

USER SYMBOLS:

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APPROVED 12 JAN 68 (A) REVISED 28 SEP 73 (B) 15 APR 74

(B) ENTIRE STANDARD REVISED

FOR NOTES SEE SHEET 2.

P.A. NAVY - AS Other Cust USAF - 11 ARMY - AV	INTERNATIONAL INTEREST **	TITLE: NUT, SELF-LOCKING, CASTELLATED, HEXAGON, COUNTERBORED, ASSEMBLED WASHER, 250°F, NON-METALLIC INSERT (FOR SELF RETAINING BOLTS)	MILITARY STANDARD MS21224
PROCUREMENT SPECIFICATION *MIL-N-25027		SUPERSEDES: MS21224(ASG)	SHEET 1 OF 2

DD FORM 1 MAR 72 672-1 (Coordinated)

PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

PLATE NO. 23055

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(a) ULTIMATE AXIAL STRENGTHS ARE NOT IN ACCORDANCE WITH MIL-N-25027.

(b) NUTS WILL ASSEMBLE ON EITHER MIL-S-7742 OR MIL-S-8879 THREADED BOLTS.

(c) CONCENTRICITY: "E" TO BE CONTRIC TO P.D. OF THREAD WITHIN .005 TIR.

(d) WASHER I.D. MAY BE .005 LESS THAN "E" MIN.

NOTES:

1. MATERIAL: a. BODY: STEEL (SEE PROCUREMENT SPECIFICATION.); CRES (A286)
   b. WASHER: ALUMINUM ALLOY, 2024-T4 OR 7075-T6.
   c. INSERT: NYLON OR EQUIVALENT
2. PROTECTIVE TREATMENT: a. NUT-CADMIUM PLATING IN ACCORDANCE WITH QQ-P-416, TYPE II, CLASS 2. PARTS WITH CLASS 3 PLATING MAY BE FURNISHED FROM SUPPLIER'S STOCK UNTIL JUNE 1975.
   b. WASHER-ALODINE MIL-C-5541.
3. THREADED ELEMENT SHALL ROTATE FREELY IN AND AGAINST THE WASHER.
* 4. PERFORMANCE: PER MIL-N-25027 EXCEPT FOR:
   VIBRATION TIME FOR NUT SIZES .3125 AND LARGER SHALL BE 120 MINUTES MINIMUM.
   LOCKING TORQUE VALUES APPLICABLE FOR ONLY ONE CYCLE AT ROOM TEMPERATURE.
   SEE (a) AND COLUMNS X, Y & Z OF TABLE.
5. SURFACE TEXTURE IN ACCORDANCE WITH ANSI-B46.1 SHALL NOT EXCEED 125 MICROINCHES.
6. BREAK ALL SHARP EDGES AND REMOVE BURRS AND SLIVERS.
7. DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCE: DECIMALS ±.010.
8. DESIGN INFORMATION: THESE NUTS ARE FOR USE WITH SELF-RETAINING BOLTS AND MAY BE USED AT TEMPERATURES 250°F AND BELOW. FOR DESIGN USAGE LIMITATIONS SEE MS33602. NUTS ARE TO BE INSTALLED IN ACCORDANCE WITH MS33540.
9. QUALIFIED PRODUCTS: ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL-25027 SHALL BE USED FOR DESIGN.
10. EXAMPLE OF PART NUMBER: MS21224-4 = .2500-28 STEEL, CADMIUM PLATED, ASSEMBLED WASHER, ALUMINUM, ALODINE TREATED, INSERT, NON-METALLIC.
△ 11. ADD "C" AFTER DASH NUMBER TO INDICATE CRES.
12. MAGNETIC PERMEABILITY OF CRES NUTS SHALL BE LESS THAN 2.0 (AIR = 1.0) FOR A FIELD STRENGTH H = 200 OERSTEDS USING A MAGNETIC PERMEABILITY INDICATOR IN ACCORDANCE WITH MIL-I-17214 OR EQUAL.

** CERTAIN PROVISIONS FOR ACROSS FLAT DIMENSIONS OF THIS STANDARD ARE THE SUBJECT OF INTERNATIONAL STANDARDIZATION AGREEMENT (ABC AIR STD 17/2). WHEN REVISION, OR CANCELLATION OF THIS STANDARD IS PROPOSED WHICH WILL AFFECT OR VIOLATE THE INTERNATIONAL AGREEMENT CONCERNED, THE PREPARING ACTIVITY WILL TAKE APPROPRIATE RECONCILIATION ACTION THROUGH INTERNATIONAL STANDARDIZATION CHANNELS, INCLUDING DEPARTMENTAL STANDARDIZATION OFFICES, IF REQUIRED.

(B) 13. PARTS WITHOUT MFR IDENT MAY BE FURNISHED FROM SUPPLIER'S STOCK UNTIL DEC 31, 1975.

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS.

APPROVED 12 JAN 68 REVISED (B) SEE SHEETS 1 & 2

P.A. NAVY-AS Other Cust USAF-11 ARMY-AV	INTERNATIONAL INTEREST **	TITLE: NUT, SELF-LOCKING, CASTELLATED, HEXAGON, COUNTERBORED ASSEMBLED WASHER, 250°F, NON-METALLIC INSERT (FOR SELF RETAINING BOLTS)	MILITARY STANDARD MS21224
PROCUREMENT SPECIFICATION * MIL-N-25027		SUPERSEDES: MS21224 (ASG)	SHEET 2 OF 2

DD FORM 1 MAR 72 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE PLATE NO 23065

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REVIEWER SYMBOLS:
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USAF - 11
ARMY - AV, EL
DSA - IS

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DOCUMENT IDENTIFIER AND TITLE MS21224B NUT, SELF-LOCKING, CASTELLATED, HEXAGON, COUNTERBORED, ASSEMBLED WASHER, 250°F, NONMETALLIC INSERT (FOR SELF RETAINING BOLTS)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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